

INDUSTRY INTERNSHIP SUMMARY REPORT

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number : 2410030084

Name of Student: SANYARITU BHATIA

Batch: 2024-28

2026-27

CANDIDATE'S DECLARATION

I, Sanyaritu Bhatia do hereby declare that the internship report titled **“Titanic Survival Prediction”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature Sanya
Student Name Sanyaritu Bhatia
Roll No. 2410030084

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature Sanya

Student Name Sanyaritu Bhatia

Roll No. 2410030084

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Internship Completion Certificate

C.ID: 214fa5a



CERTIFICATE

OF COMPLETION
PROUDLY PRESENTED TO

Sanyaritu Bhatia

has successfully completed 4 weeks of a virtual internship program in
Data Science

with wonderful remarks at **CODSOFT** from 01/08/2026 to 31/08/2026.

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.



Founder



contact@codsoft.in

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Date: 01/09/2026

4. Titanic Survival Prediction

4.1 Introduction

The “Titanic Survival Prediction” project is a data science initiative developed during the virtual internship at CodSoft. The sinking of the RMS Titanic is one of the most infamous shipwrecks in history. While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others. The primary objective of this project was to build a machine learning model capable of predicting whether a passenger survived or not, based on specific parameters such as age, gender, ticket class, fare, cabin, and port of embarkation. The methodology involved extensive data preprocessing, including handling missing values and categorical variable encoding, followed by Exploratory Data Analysis (EDA) to identify trends. A Logistic Regression classifier was trained to discern patterns between passenger attributes and survival rates. This report details the technical implementation, system design, and the learning outcomes achieved during the internship period.

4.2 Organization Profile

CodSoft is a technology and software development organization that offers internship and learning opportunities to students and budding professionals. The organization aims at enabling students to gain first-hand exposure to the technology field. It allows students to develop technical expertise through acquiring industry-ready technological skills.

The organization offers virtual internship programs in different areas, including Web Development, Python Programming, Java Programming, Data Science, Machine Learning, and other trending fields. The training provided is practical and targets enabling participants to integrate their acquired skills into their career development. Virtual internships enable students to further expand their knowledge and skill set while dealing with projects and assignments.

Moreover, CodSoft fosters a project-based learning model where participants are required to complete a list of tasks and projects within a specified time frame. Consequently, participants are able to grow on their own, develop their skills further, and become more proficient in programming and other related fields. Additionally, the virtual aspect of the training allows participants to operate from the comfort of their homes while still being able to comply with all the regulations and requirements of the projects and assignments provided.

CodSoft virtual internships enable students to bridge the gap between theoretical knowledge and practical experience. It provides participants with the required technical expertise, self-learning and -practice, and professional development to become better and more competitive in their careers. In essence, the organization allows students and budding professionals to expand their knowledge and skill base through gaining exposure to the actual field of work in the comfort of their homes.

4.3 Problem Statement

The problem requires analysing the Titanic passenger dataset to identify patterns that correlate with survival rates. The dataset contains features like 'Pclass' (Ticket Class), 'Sex', 'Age', and 'Fare'. The research objective is to create a binary classification model where the target variable is 'Survived' (0 = Deceased, 1 = Survived). We aim to maximize the accuracy of these predictions on unseen data.

4.4 Problem Objectives

The main objective of the Titanic Survival Prediction project is to apply data science techniques to a real-world dataset to predict binary outcomes. Specifically, the goals are:

- To clean and preprocess raw data containing null values and inconsistent formatting.
- To perform Exploratory Data Analysis (EDA) to visualize survival correlations with Age, Gender, and Class.
- To build and train a predictive model using machine learning algorithms (Logistic Regression).

4.5 Scope of the Project

This project falls under the domain of **Data Science and Machine Learning**. It utilizes Supervised Learning techniques. Supervised learning involves training a model on a labelled dataset, where the correct output is known. The specific task is Binary Classification. The tools used include Python for programming and libraries such as Pandas for data manipulation, Matplotlib/Seaborn for visualization, and Scikit-Learn for modelling.

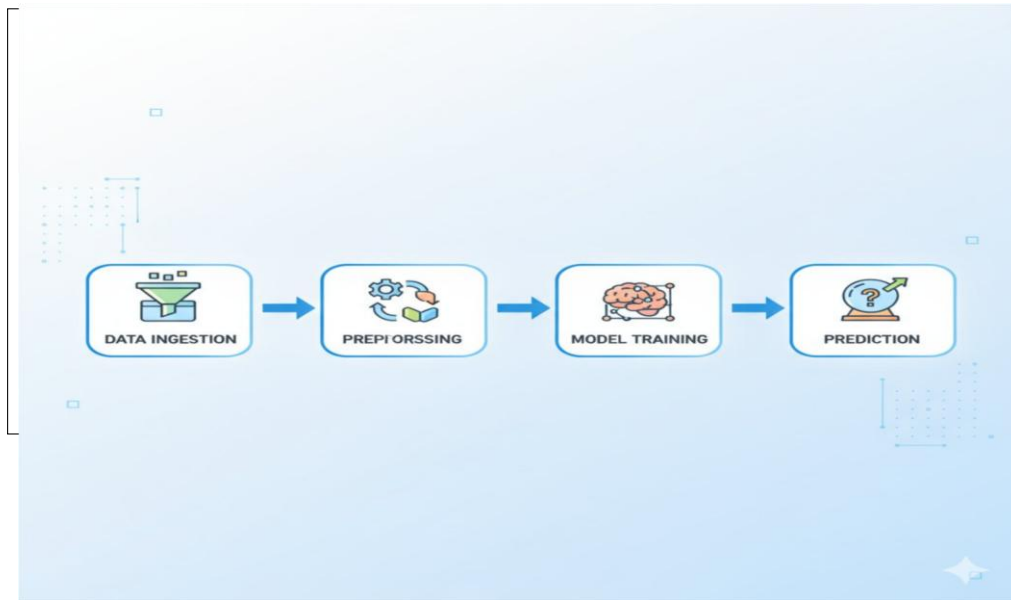
4.6 Technologies and Tools Used

- **Jupyter Notebook:** The Integrated Development Environment (IDE) used for writing and executing code interactively. It allows for immediate visualization of data plots alongside code.
- **Pandas:** A powerful library used for data manipulation. It was used to load the Titanic CSV file into a DataFrame, handle missing values (imputation), and perform feature selection.
- **NumPy:** Used for efficient numerical operations and array handling, which is essential for the mathematical calculations underlying machine learning algorithms.
- **Matplotlib & Seaborn:** Visualization libraries used to create bar charts, histograms, and heatmaps. These were crucial for the EDA phase to understand the distribution of data (e.g., survival rates by Gender or Class).

- **Scikit-Learn:** The primary machine learning library used to implement the classification algorithms (Logistic Regression) and metric functions (Accuracy Score, Confusion Matrix) for validation.

4.7 System Architecture

The system follows a standard Data Science pipeline architecture, consisting of distinct phases that transform raw data into actionable predictions.



4.8 Methodology

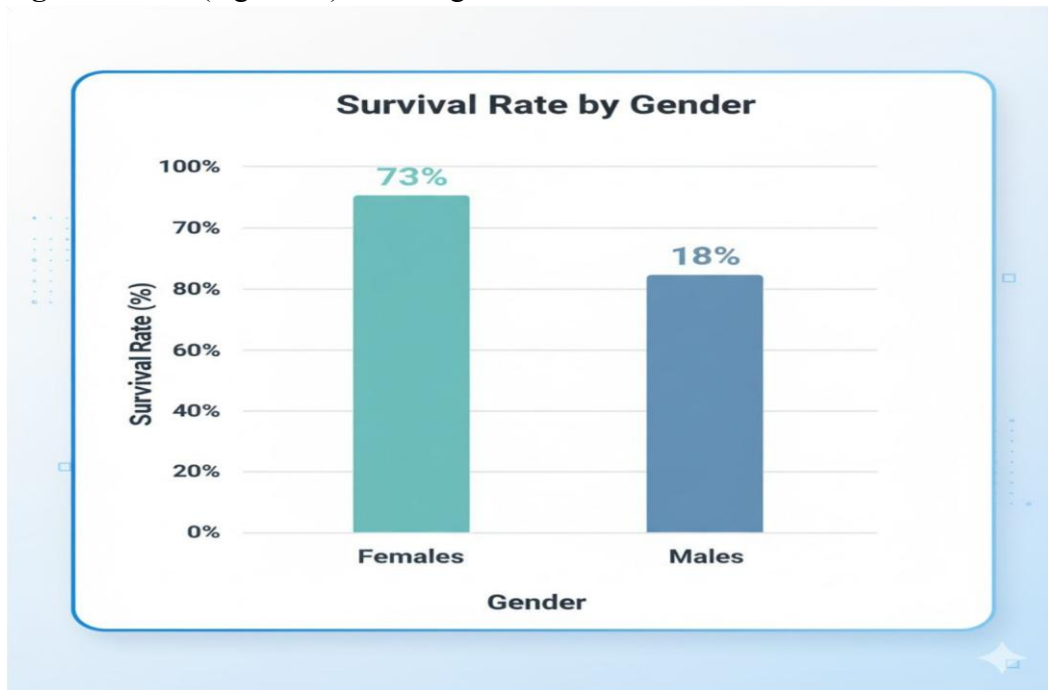
The methodology adopted for this project is based on the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework:

1. Business Understanding: Determining that the goal is to predict survival.
2. Data Understanding: Visualizing the dataset to find correlations (e.g., Women and children first).
3. DataPreparation: Cleaning the dataset and handling outliers.
4. Modeling: Applying Logistic Regression.
5. Evaluation: Checking the accuracy score against a validation set.

4.9 Expected Outcomes

The Exploratory Data Analysis yielded significant insights into the historical event:

- **Gender:** Female passengers had a significantly higher survival rate compared to males. This confirms the “Women and children first” protocol.
- **Class:** Passengers in 1st Class (Pclass 1) had a higher chance of survival than those in 3rd Class, likely due to cabin proximity to the deck.
- **Age:** Children (Age < 10) had a higher survival rate.



The Logistic Regression model was evaluated using the accuracy metric.

- **Accuracy:** The model achieved an accuracy of approximately **80%** on the test data. This means the model correctly predicted the survival status for 8 out of 10 passengers in the test set.
- **Confusion Matrix:** The matrix showed a balanced number of False Positives and False Negatives, indicating a stable model.

4.10 Offer Letter and Completion certificate

 **CodSoft**

INTERNSHIP OFFER LETTER

Date : 29/07/2026

ID: BY26RY225938

Dear **Sanyaritu Bhatia**,

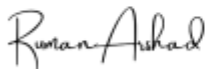
We would like to congratulate you on being selected for the “**Data Science**” virtual internship position with “**CodSoft**”. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **01 August 2026 to 31 August 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Sincerely



Ruman Arshad
Program Manager


ISO Number - TSNUK79053


MSME Regd



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Date: 01/09/2026

5. Bibliography / References

- 1) Codsoft – Virtual Internship Programs.
- 2) Kaggle. Titanic – Machine Learning from Disaster: Titanic Dataset.
- 3) Python Software Foundation. Python Documentation.
- 4) Pandas Development Team. Pandas Documentation and User Guide.
- 5) Project Jupyter. Jupyter Notebook Documentation and Resources.
- 6) Scikit-Learn Developers. Scikit-Learn User Guide – Machine Learning in Python.
- 7) Matplotlib Development Team. Matplotlib Documentation.
- 8) Seaborn Development Team. Seaborn Statistical Data Visualization.